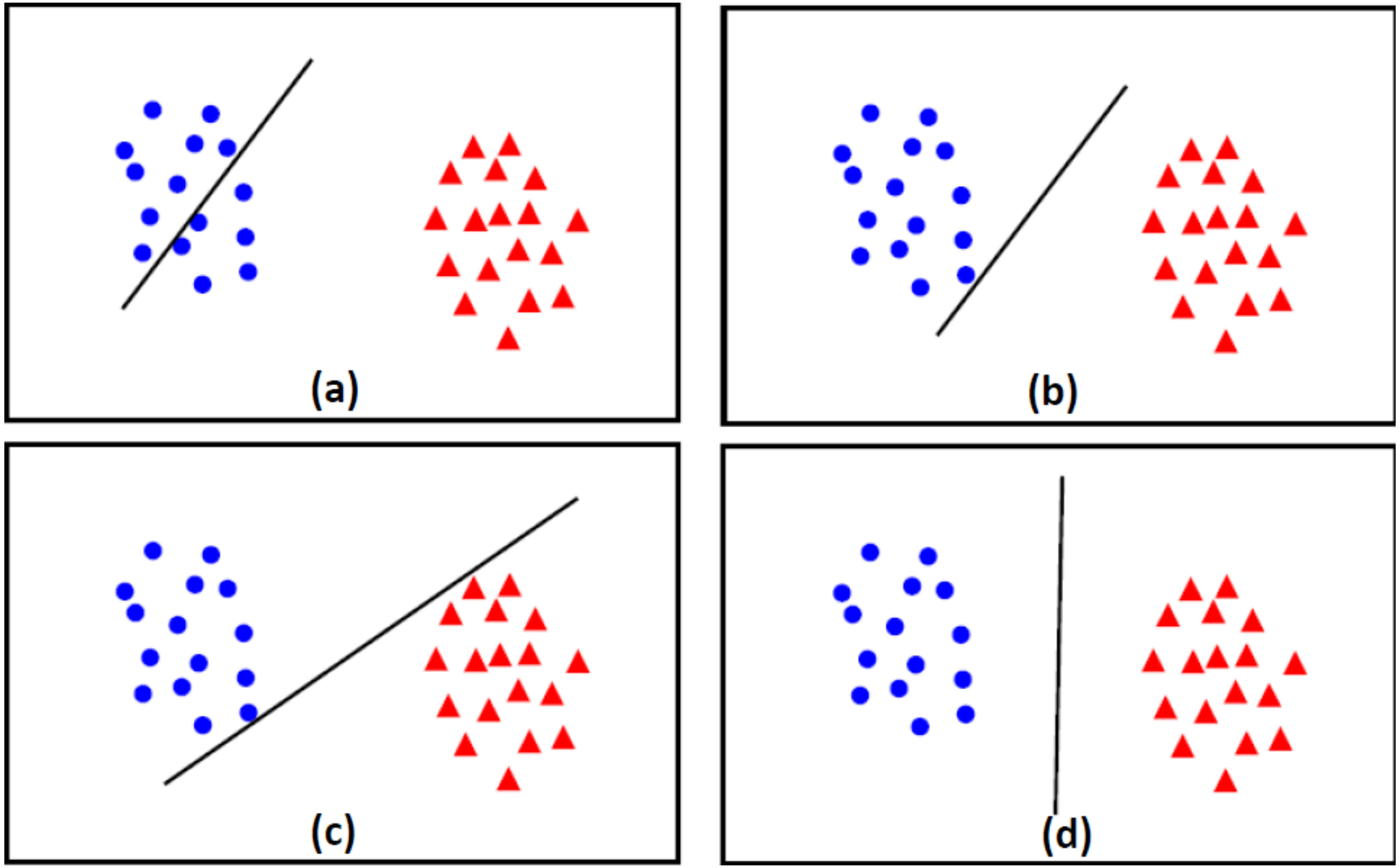
Support Vector Machine (SVM)

What are SVMs?

A support vector machine (SVM) is a supervised machine learning algorithm used for classification and regression tasks, but it is most commonly used for classification.

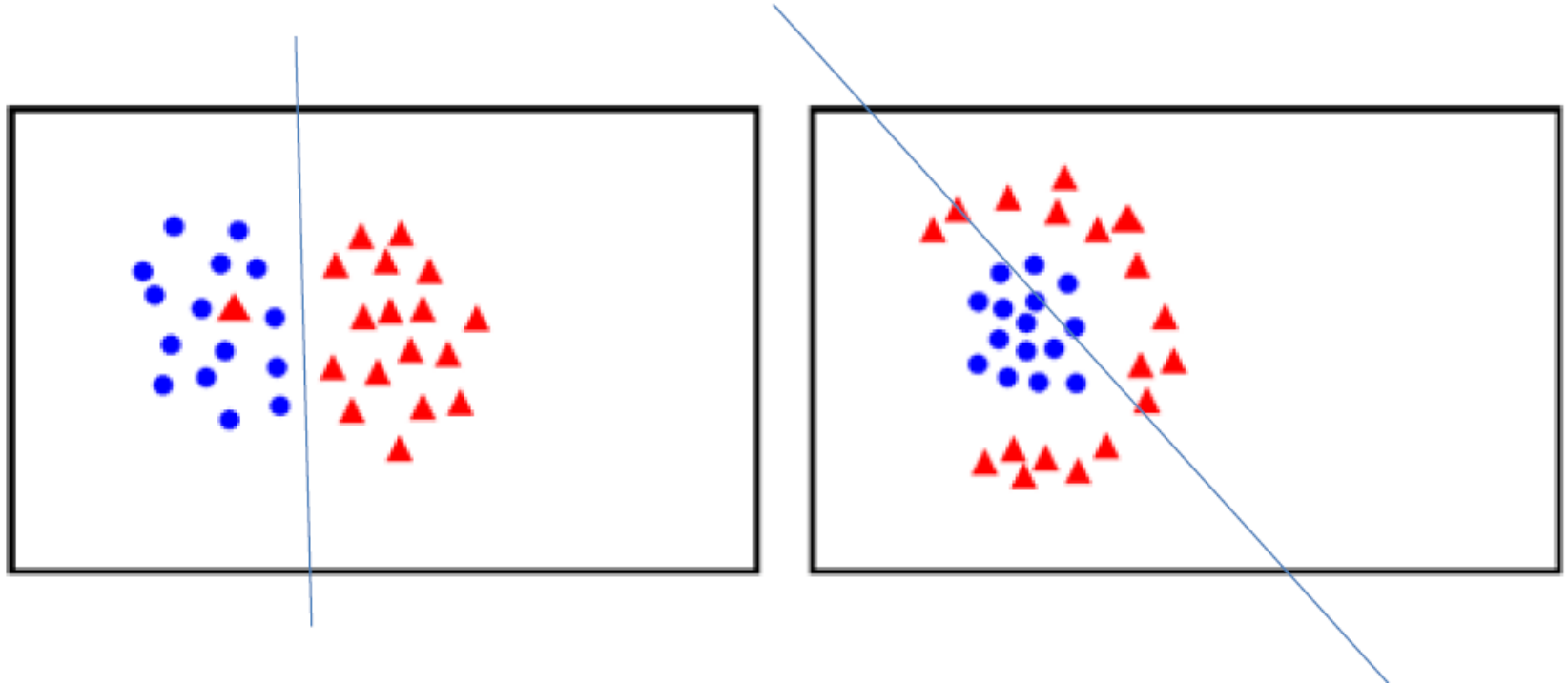
SVMs were developed in the 1990s by Vladimir N. Vapnik and his colleagues, and they published this work in a paper titled "Support Vector Method for Function Approximation, Regression Estimation, and Signal Processing" in 1995.

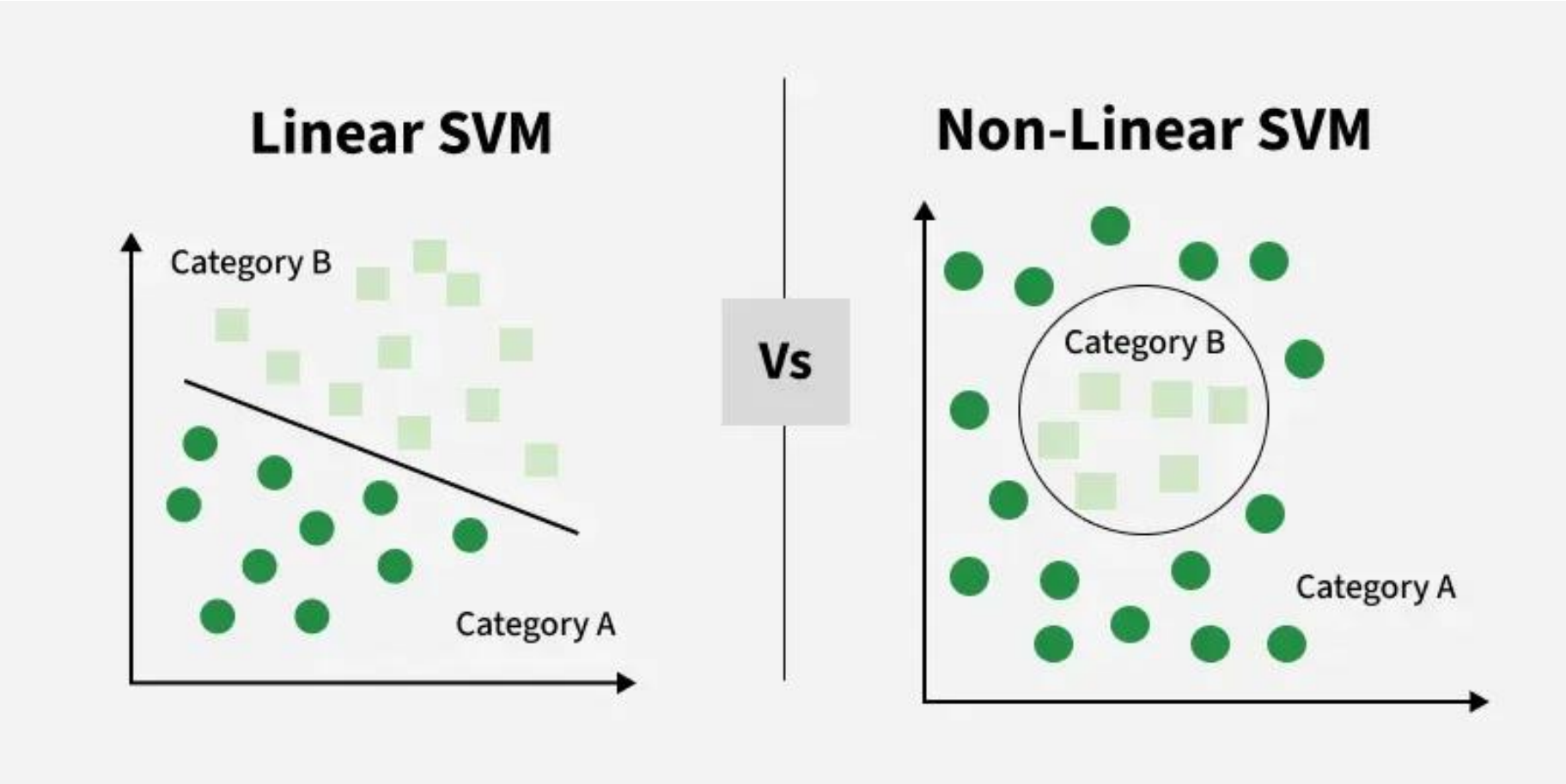
In classification, SVM classifies data by finding an optimal line or hyperplane that best separates data points of different classes in a feature space.



Linear Problem

Non-Linear Problem

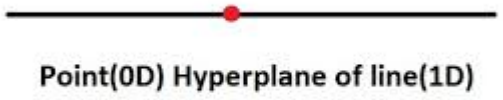



Linear SVM
Category B
Category A
Vs
Non-Linear SVM
Category B
Category A

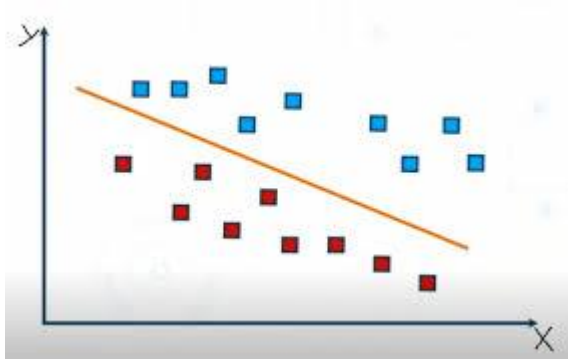
Linear SVM

By its nature, the **hyperplane** separates the space into two half-spaces.

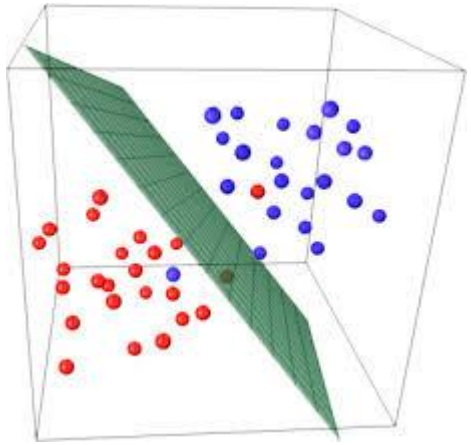
As an example:
- A **point** is a hyperplane in **1-dimensional** space,
- A **line** is a hyperplane in **2-dimensional** space,
- And a **plane** is a hyperplane in **3-dimensional** space.



1-dimensional space



2-dimensional space

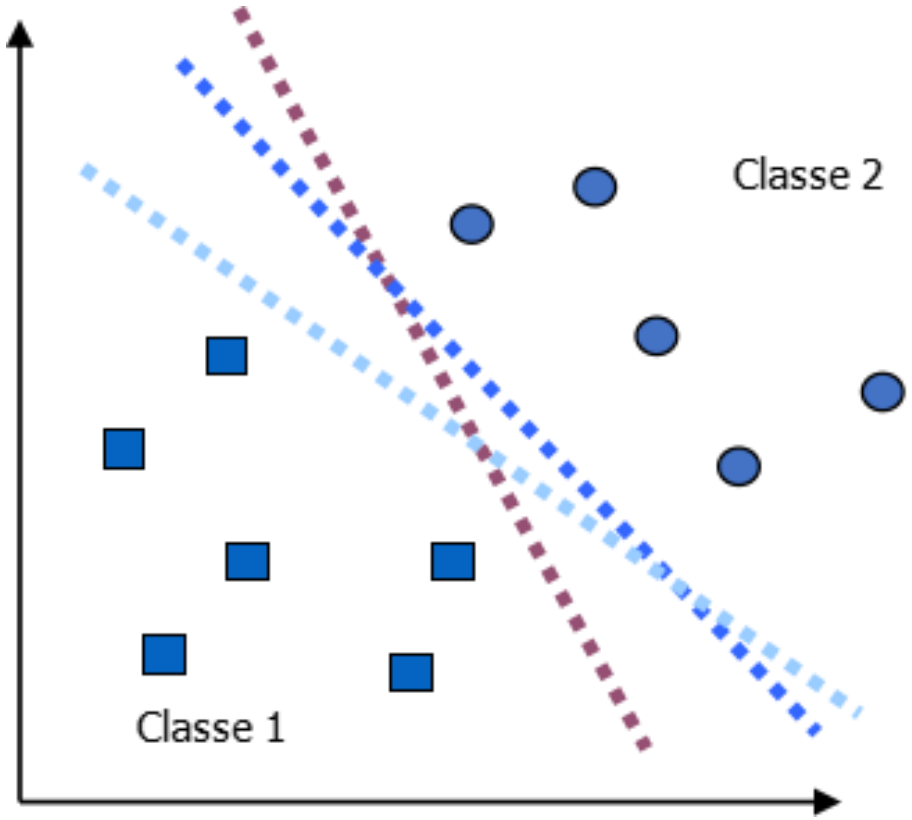


3-dimensional space

How does Support Vector Machine Algorithm Work?

SVM is a Machine Learning technique of classification based on the use of the Linear Discriminant Function $f(X) = w^T X + w_0$, which represents a hyperplane in the feature space.

A discriminant function represents a surface which separates the patterns so that the patterns from the two classes lie on the opposite sides of the surface.



The challenge

The challenge is to classify given data points using a linear discriminant function in order to minimize the error rate.

There are infinite numbers of answers possible, but which is optimal and best?

$$f(x) = w_0 + \sum_{d=1}^{D} w_d x_d$$

$$= w_0 + \mathbf{w}^T x$$

The variables used are :

- w : The weight vector(defining the orientation of the hyperplane).
- x: Input feature vectors.
- w_0 or b: bias term (defining the position of the hyperplane)

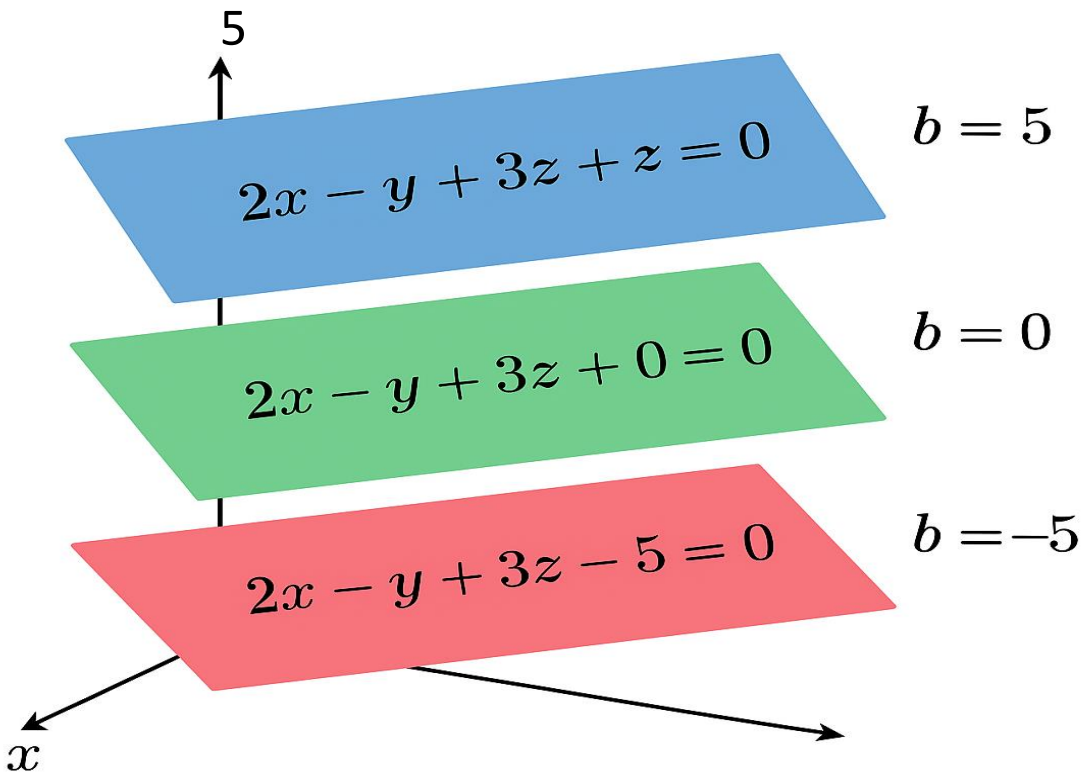
For example:

In 2-Dim, the equation of the hyperplane is : $w_1x+w_2y+w_0=0$

In 3-Dim, the equation of the hyperplane is : $w_1x+w_2y+w_3z+b=0$

Etc....

In 3-Dim, the equation of the hyperplane is : $w_1x+w_2y+w_3z+b=0$



Here, **w** =(2,-1,3) represents the normal to the hyperplane and **b** € R is the offset (Decalage) to y-axis (or b is the bias (intercept).

There are infinite numbers of answers possible, but which is optimal and best?

The linear discriminant function with maximum margin is the optimal and best solution to the above question and is obtained by SVM by determining a separating hyperplane which is optimal according to a criterion as follows:

Suppose Class labels are denoted by +1 and -1. The SVM determines the optimal linear discriminant function with the help of **support vectors** and is given by $w^TX + b = 0$, where $Y_i = 0$ for all patterns i. This is the equation of a hyperplane.

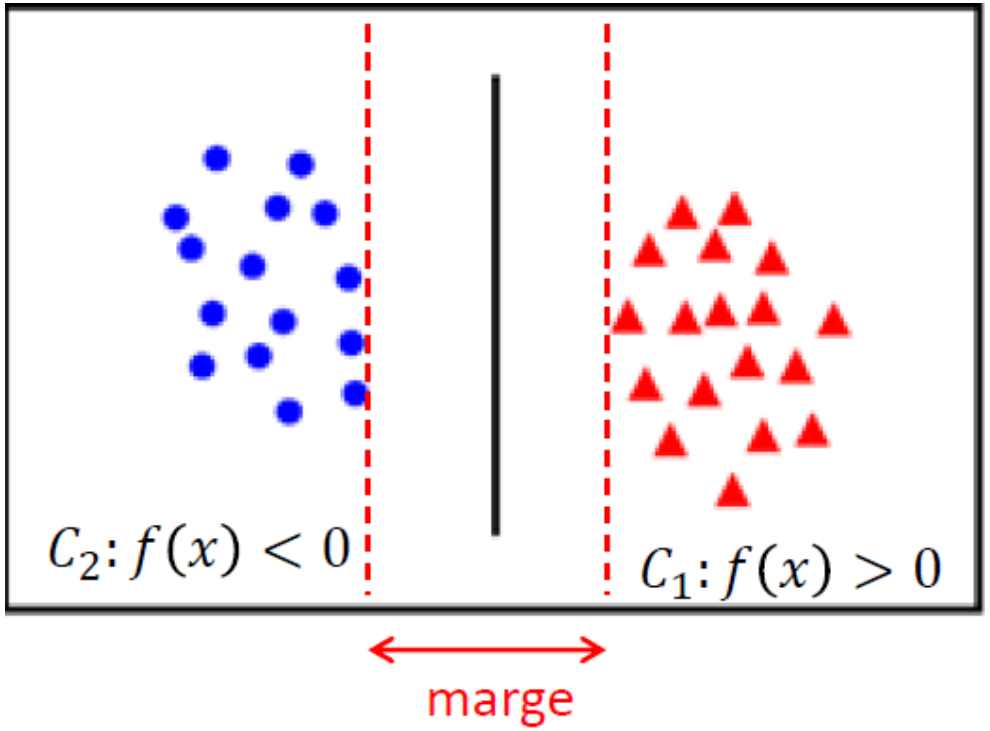
The goal is to **maximize** the **margin** (i.e. the distance between the hyperplane and the nearest data points from each class). These points are called **support vectors, <u>which help the model generalize better and avoid overfitting</u>**.

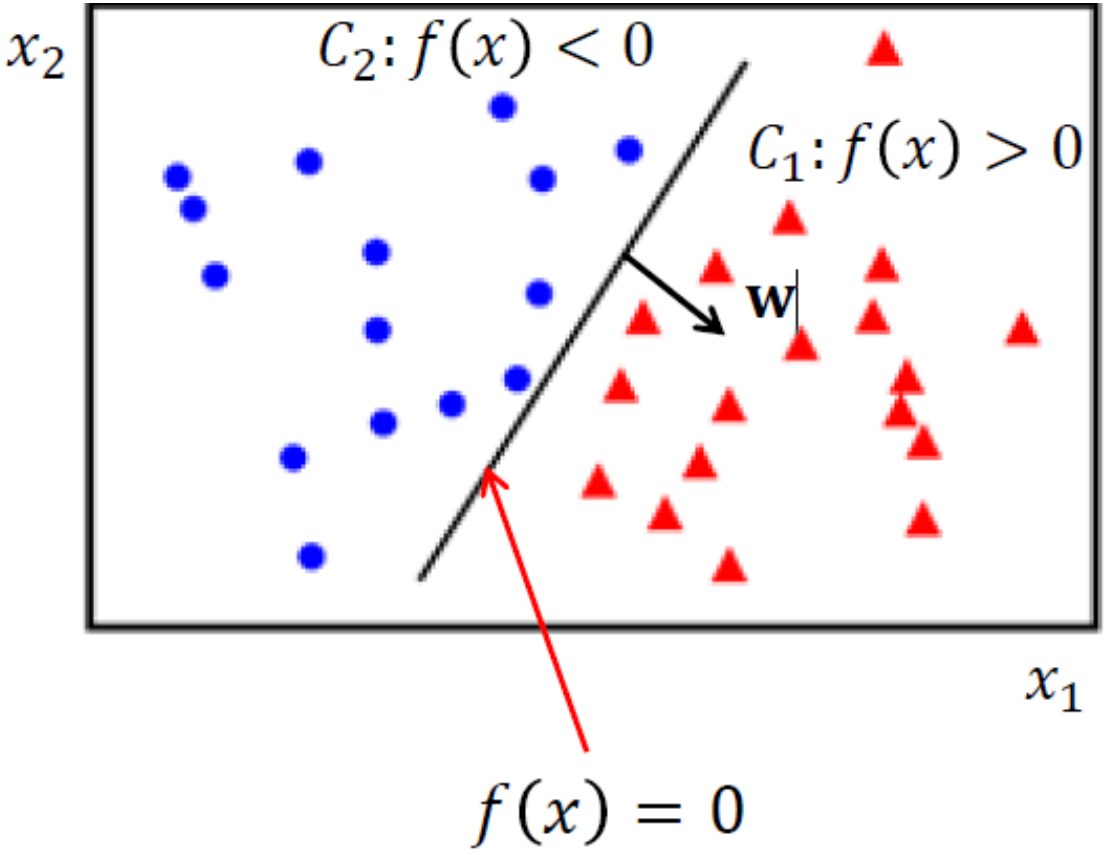
To find the maximum margin between two classes, two support planes are determined.

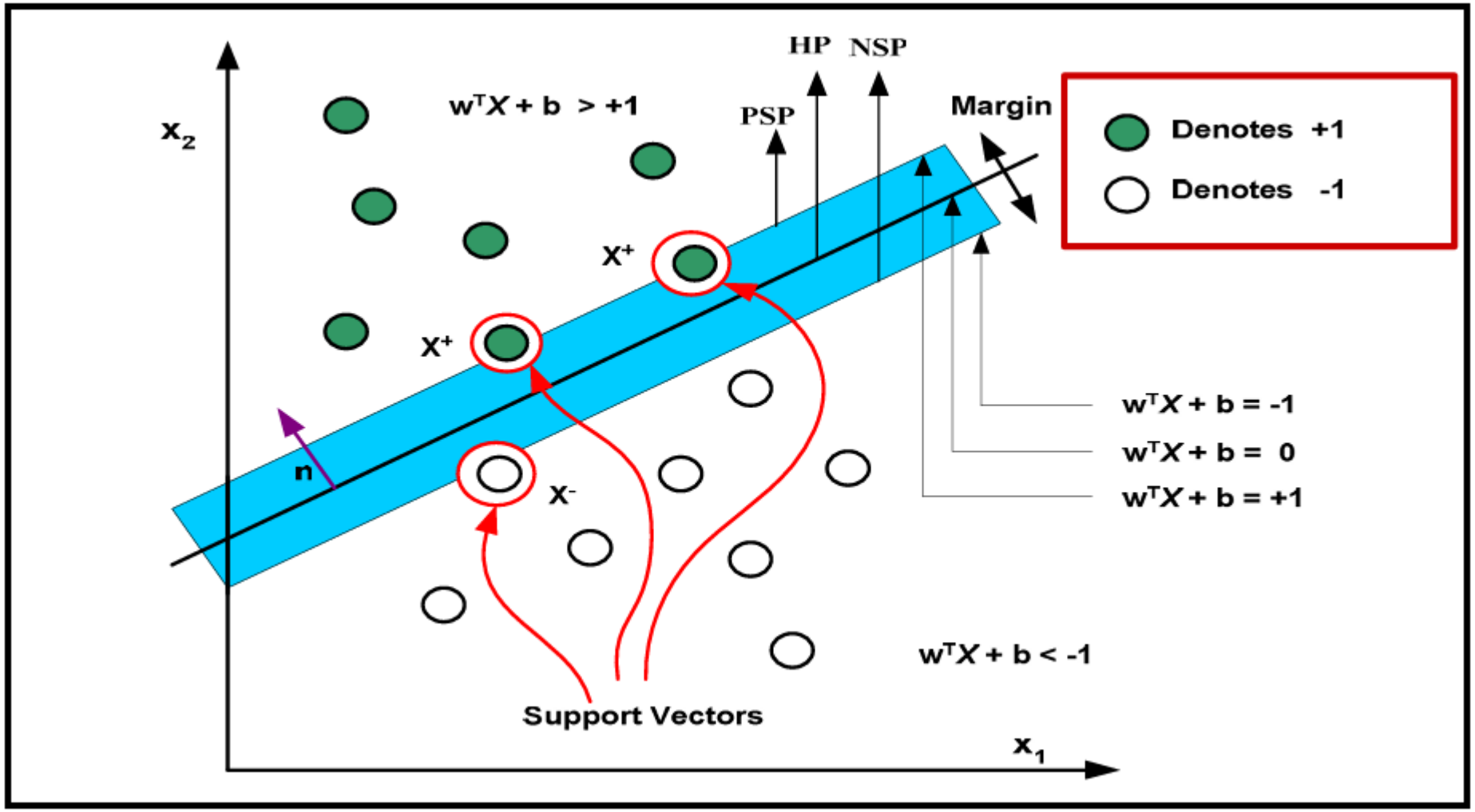
Positive class support plane is denoted by $w^Tx + b = +1$ which lies in positive class(= +1).
Negative class support plane is denoted by $w^Tx + b = -1$ which lies in negative class(= -1).

The best hyperplane also known as the **"hard margin".**

Here, **w** =(w_1,w_2) represents the normal to the hyperplane and **b** € R is the offset (Decalage) to y-axis (or b is the bias (intercept).

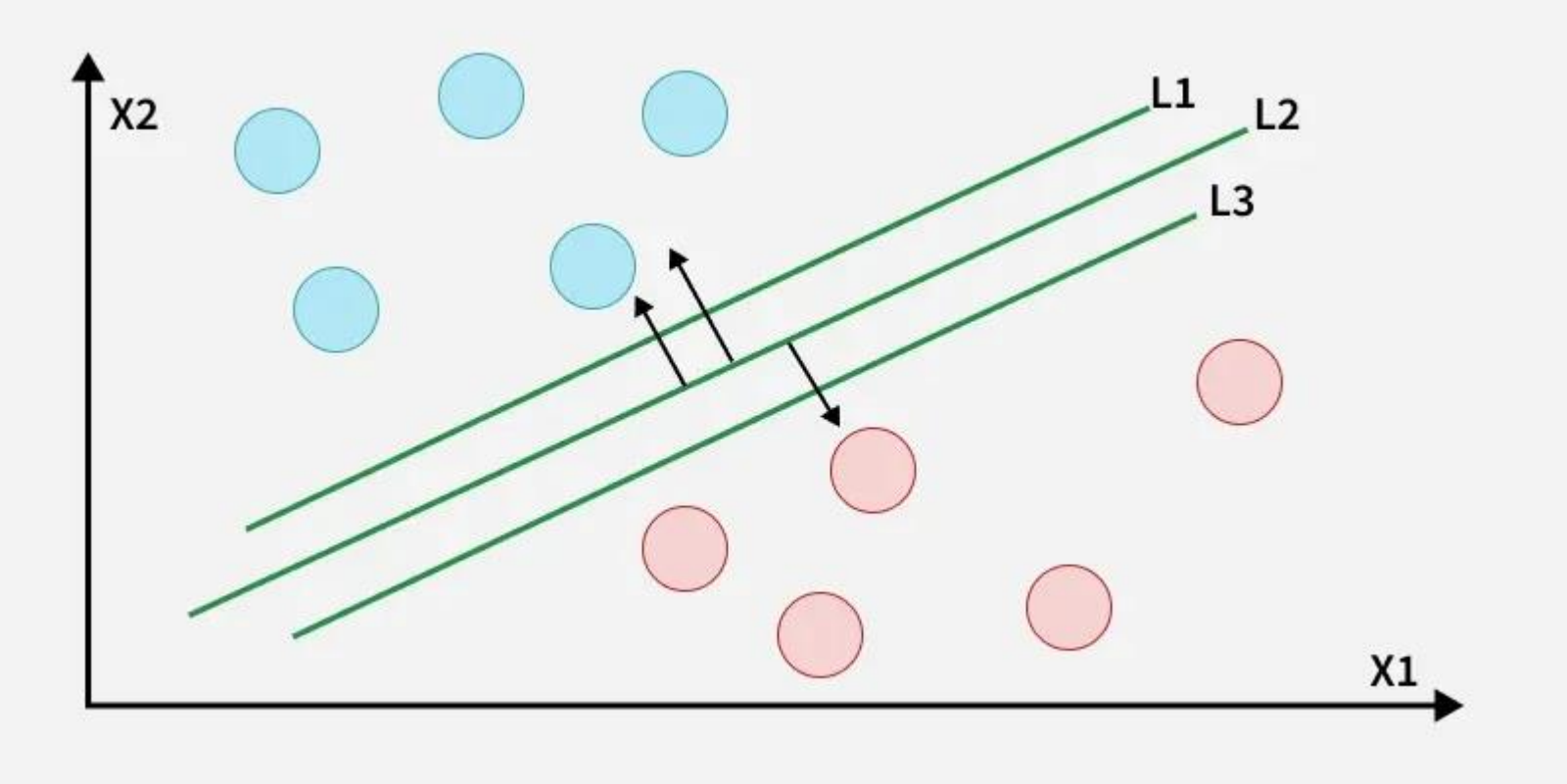





HP
NSP
PSP
Margin
$w^T X + b > +1$
x_2
x_1
X^+
X^+
X^-
n
Denotes +1
Denotes -1
$w^T X + b = -1$
$w^T X + b = 0$
$w^T X + b = +1$
$w^T X + b < -1$
Support Vectors

How does Support Vector Machine Algorithm Work?

The key idea behind the SVM algorithm is to find the hyperplane that best separates two classes by maximizing the margin between them. This margin is the distance from the hyperplane to the nearest data points (support vectors) on each side.



The best hyperplane also known as the **"hard margin"** is the one that maximizes the distance between the hyperplane and the nearest data points from both classes. This ensures a clear separation between the classes. So from the above figure, we choose L2 as hard margin. Let's consider a scenario like shown below:

The **hyperplane parameters (w** and **b)** should be such that the line should pass through exactly in the middle of the space between two classes and the perpendicular distance of nearest points (**support vectors**) of two classes to the hyperplane must be same.

In SVMs we are trying to find a **decision boundary** (hyperplane) that maximizes the "margin" separating the positives from the negative training data points.

The **distance between two parallel hyperplanes** is the **margin** width **M=** $\frac{2}{\|\vec{w}\|}$
.

- $\|w\|$ = the **norm (length)** of the weight vector w.
- The **larger $\|w\|$** → the **smaller the margin**.
- The **smaller $\|w\|$** → the **wider the margin**.

Our goal is to minimise the $\|w\|$

To find this **decision boundary** (hyperplane), we minimize: $\frac{1}{2}\|\vec{w}\|^2$ subject to the **<u>constraints</u>** $y^{(i)}(\vec{w}\vec{x}_i + b) - 1 \geq 0 \quad \forall i$

$$\begin{cases} \vec{w}\vec{x}_2 + b \geq 1 \text{ pour } y^{(+1)} = 1 \\ \vec{w}\vec{x}_1 + b \leq -1 \text{ pour } y^{(-1)} = -1 \end{cases} \longrightarrow \begin{cases} y^{(+1)}.(\vec{w}\vec{x}_2 + b) \geq 1 \text{ pour } y^{(+1)} = 1 \\ y^{(-1)}.(\vec{w}\vec{x}_1 + b) \geq 1 \text{ pour } y^{(-1)} = -1 \end{cases}$$

$$\Downarrow$$

$$y^{(i)}(\vec{w}\vec{x}_i + b) \geq 1 \qquad \forall i$$

Optimization problem with constraints:

Lagrangian optimization is a method for solving optimization problems with constraints.

The resulting **Lagrange multiplier** equation we try to optimize is:

$$Q = \frac{1}{2}\|\vec{w}\|^2 - \sum_{i=1}^{n} \alpha_i(y^{(i)}(\vec{w}\vec{x}_i + \mathrm{b}) - 1)$$

such that $\alpha_i \neq 0$

Solving the above Lagrangian optimization problem will give us **w**, **b**, and **alphas**. These parameters determine a unique maximal margin solution.

The α_i are **Lagrange multipliers**. The function Q must be minimized on **w** and **b** and maximized on the α_i.

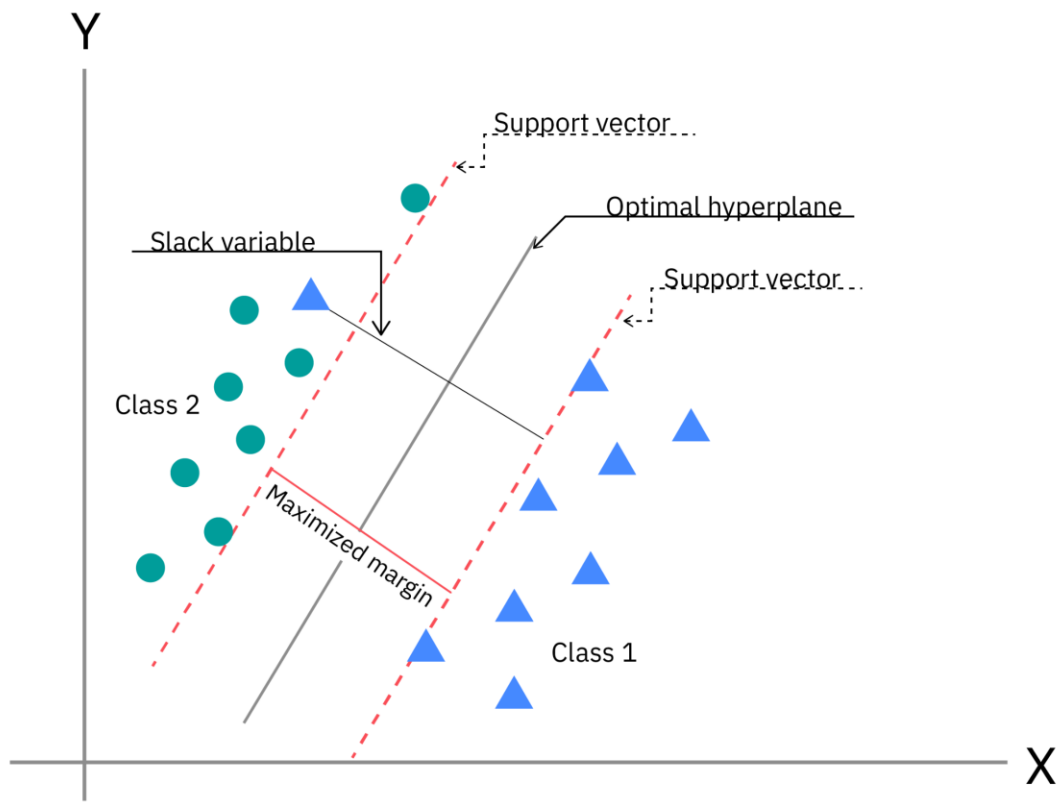
Using the derivatives on **w** and on **b:** $\frac{dQ}{dw} = 0 \quad et \quad \frac{dQ}{db} = 0$

$$Q = \frac{1}{2}\|\vec{w}\|^2 - \sum_{i=1}^{n} \alpha_i (y^{(i)} (\vec{w}\vec{x}_i + \mathrm{b}) - 1)$$

We will have:

$$\frac{dQ}{dw} = \vec{w} - \sum_{i=1}^{n} \alpha_i \, y^{(i)} \vec{x}_i = 0 \longrightarrow \vec{w} = \sum_{i=1}^{n} \alpha_i \, y^{(i)} \vec{x}_i$$

$$\frac{dQ}{db} = 0 \longrightarrow \sum_{i=1}^{n} \alpha_i \, y^{(i)} = 0$$


Y
Support vector
Optimal hyperplane
Slack variable
Support vector
Class 2
Maximized margin
Class 1
X

Key Concepts of Support Vector Machine

•**Hyperplane**: A decision boundary separating different classes in feature space and is represented by the equation wx + b = 0 in linear classification.

•**Support Vectors**: The closest data points to the hyperplane, crucial for determining the hyperplane and margin in SVM.

•**Margin**: The distance between the hyperplane and the support vectors. SVM aims to maximize this margin for better classification performance.

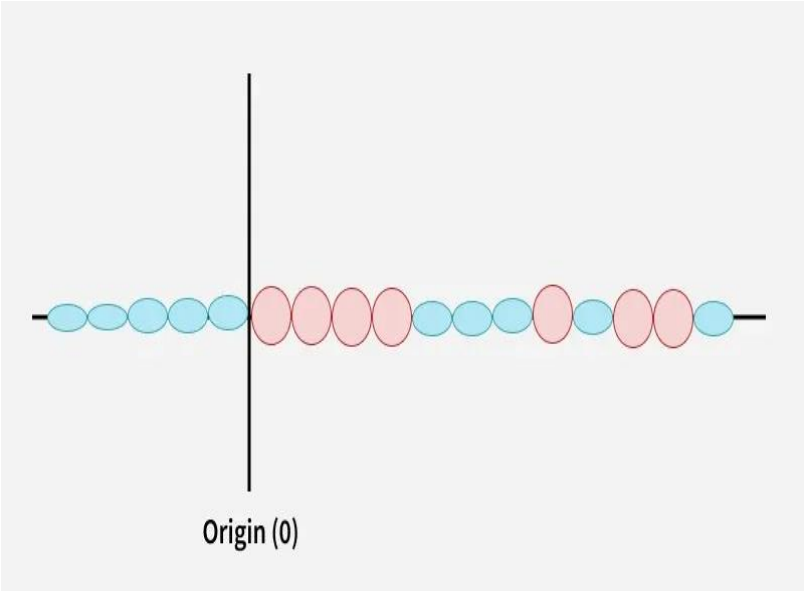
•**Kernel**: A function that maps data to a higher-dimensional space enabling SVM to handle non-linearly separable data.

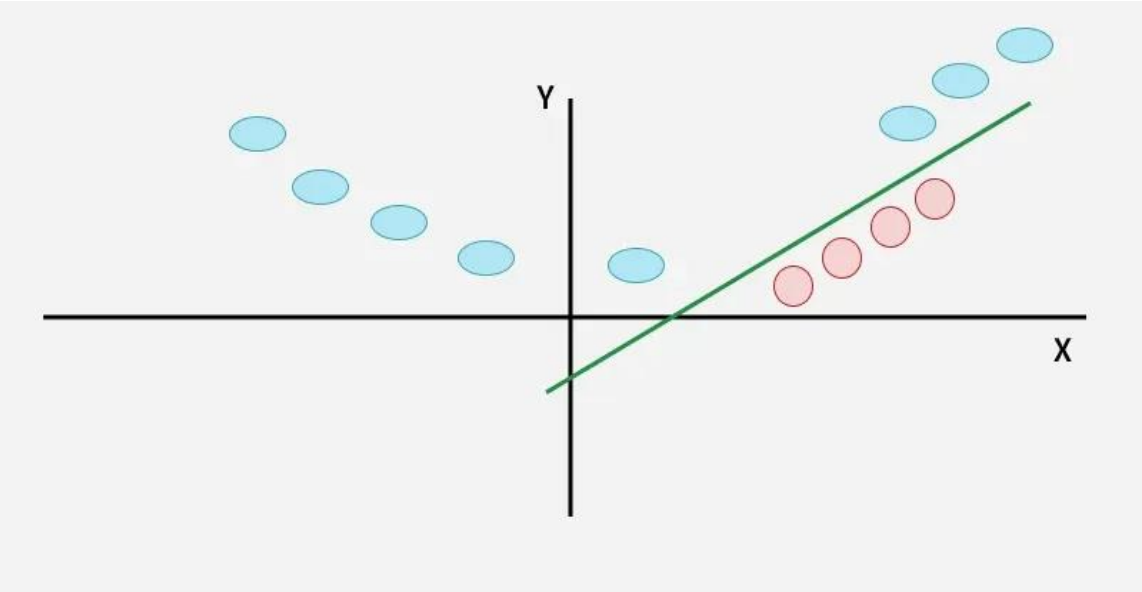
•**Hard Margin**: A maximum-margin hyperplane that perfectly separates the data without misclassifications.

What if data is not linearly separable?

When data is not linearly separable i.e it can't be divided by a straight line, SVM uses a technique called kernels to map the data into a higher-dimensional space where it becomes separable. This transformation helps SVM find a decision boundary even for non-linear data.

For example, consider data points that are not linearly separable. By applying a kernel function, SVM transforms the data points into a higher-dimensional space where they become linearly separable.





In this case, the new variable y is created as a function of distance from the origin.